OptiWealth - Comprehensive Feature Audit

Generated: 2026-01-25
Application: OptiWealth - AI-Powered Portfolio Optimization Platform

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✅ Implemented Features

🔐 Authentication & Security

Feature	Status	Description
User Registration	✅ Complete	Email-based signup with password strength validation
Secure Login	✅ Complete	JWT-based auth with access + refresh tokens
Multi-Factor Auth (MFA)	✅ Complete	Email OTP verification for sensitive logins
Captcha Protection	✅ Complete	Math-based captcha after failed attempts
Password Reset	✅ Complete	Secure token-based email reset flow
Account Deletion	✅ Complete	Permanent deletion with cascade cleanup + confirmation email
Session Management	✅ Complete	View active sessions, revoke devices remotely
Rate Limiting	✅ Complete	IP-based rate limiting with configurable thresholds
Brute Force Protection	✅ Complete	Blocks IPs after excessive failed attempts

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📊 Portfolio Management

Feature	Status	Description
Multiple Portfolios	✅ Complete	Create/edit/delete named portfolios
Holdings CRUD	✅ Complete	Add/update/remove individual holdings
Transaction History	✅ Complete	Track all BUY/SELL/DIVIDEND transactions
Real-time Pricing	✅ Complete	Live prices from Alpha Vantage API
Import from Broker	✅ Complete	Parse Excel/CSV from Zerodha, Groww, Angel One
Target Allocations	✅ Complete	Set desired % per asset for rebalancing
P&L Calculations	✅ Complete	Automatic profit/loss with cost basis

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📈 Analytics & Visualization

Feature	Status	Description
Portfolio History Chart	✅ Complete	Historical value tracking with 1M/3M/6M/1Y/ALL periods
Allocation Pie Chart	✅ Complete	Visual breakdown by asset

Performance Chart	✅ Complete	Line chart with timeline comparison
Correlation Matrix	✅ Complete	Heatmap showing asset correlations
Efficient Frontier	✅ Complete	Mean-variance optimization curve
Monte Carlo Simulation	✅ Complete	1000+ simulations for future projections
Benchmark Comparison	✅ Complete	Compare portfolio vs NIFTY 50
Risk Metrics	✅ Complete	Sharpe Ratio, Beta, Volatility, VaR

⚖️ Rebalancing

Feature	Status	Description
Drift Detection	✅ Complete	Compares actual vs target allocations
Rebalance Wizard	✅ Complete	Step-by-step UI for rebalancing decisions
Buy/Sell Suggestions	✅ Complete	Calculated trades to reach targets
Execute Rebalance	✅ Complete	One-click apply rebalancing trades
Optimization Mode	✅ Complete	Uses efficient frontier for optimal weights

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💵 Dividends

Feature	Status	Description
Dividend Tracking	✅ Complete	Record dividend payments per holding
Dividend Summary	✅ Complete	YTD, monthly breakdown, top payers
Dividend Card	✅ Complete	Dashboard component with stats

💸 Tax Tools

Feature	Status	Description
Capital Gains Calc	✅ Complete	Short-term vs long-term gains summary
Tax-Loss Harvesting	✅ Complete	Identifies holdings with unrealized losses
Tax Summary Card	✅ Complete	Dashboard component with tax estimates

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📤 Export & Reports

Feature	Status	Description
Excel Export	✅ Complete	Download portfolio as .xlsx with metrics
PDF Report	✅ Complete	Professional PDF with holdings + transactions

Export Button	✅ Complete	Frontend UI component

🤖 AI Features

Feature	Status	Description
Portfolio Analysis	✅ Complete	AI-generated insights on holdings
Recommendations	✅ Complete	Buy/Sell/Hold suggestions
Explanation Engine	✅ Complete	Detailed reasoning for each recommendation
AI Insights Card	✅ Complete	Dashboard component displaying AI advice

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🔔 Notifications

Feature	Status	Description
Email Notifications	✅ Complete	Welcome, password reset, account deleted emails
WhatsApp Alerts	✅ Complete	Via CallMeBot API integration
Telegram Bot	✅ Complete	Optional alternative to WhatsApp
Notification Settings	✅ Complete	Manage preferences in Settings page
Drift Alerts	✅ Complete	Alert when allocation drifts beyond threshold

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🎨 UI/UX

Feature	Status	Description
Responsive Design	✅ Complete	Works on mobile, tablet, desktop
Dark/Light Support	✅ Partial	Light mode complete, dark mode CSS variables exist
Loading Animations	✅ Complete	Premium AnimatedLoader component
Toast Notifications	✅ Complete	Success/error feedback via Sonner
SweetAlert Popups	✅ Complete	Premium modals for logout/delete
Skeleton Loading	✅ Complete	Placeholder animations while loading
Landing Page	✅ Complete	Marketing page with features overview

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⚠️ Partially Implemented / Needs Improvement

Feature	Status	Notes
Email Verification	⚠️ Partial	`is_verified` field exists but verification flow not fully enforced
Broker Integration	⚠️ Partial	Import works but no live broker API connection

Price Alerts	⚠️ Partial	Backend logic exists but no frontend UI to set alerts
Investment Goals	⚠️ Partial	Database model exists but no UI/API endpoints implemented
Dark Mode	⚠️ Partial	CSS variables exist but toggle switch not implemented
Session Recording	⚠️ Partial	Sessions created but not consistently tracked on every login
Report Scheduling	⚠️ Partial	Scheduler service exists but UI not connected

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❌ Not Yet Implemented

Feature	Notes
TOTP Authenticator	`totp_secret` field exists but QR code setup not implemented
Backup Codes	`backup_codes` field exists but generation/usage not implemented
IP Whitelisting	`allowed_ips` field exists but not enforced
Two-Way Broker Sync	Would require OAuth with Zerodha/Groww APIs
Real-time WebSockets	SSE/WS for live price updates
Push Notifications	Mobile push via Firebase/APNS
Mobile App	React Native or Flutter app
Watchlist	Track stocks without adding to portfolio
News Feed	Stock news integration
Social Features	Share/compare portfolios
Backtesting	Test strategies on historical data

🚀 Future Feature Suggestions

High Priority (Quick Wins)

1. **Email Verification Enforcement**
   - Require email verification before first login
   - Add resend verification email option
2. **Investment Goals UI**
   - Create goals (retirement, education, emergency fund)
   - Track progress with visual indicators
   - Project completion date based on current growth
3. **Dark Mode Toggle**
   - Add theme switcher in settings
   - CSS variables already exist, just need toggle logic

4. **Price Alerts UI**
    - Set alerts when stock hits target price
    - Receive WhatsApp/email notification
5. **Recurring Investment Tracker**
    - Track SIPs (Systematic Investment Plans)
    - Auto-calculate expected vs actual contributions

Medium Priority (Valuable Additions)

6. **Watchlist Module**
    - Track stocks without adding to portfolio
    - Quick add to portfolio from watchlist
7. **News Integration**
    - Pull headlines from financial news APIs
    - Show relevant news for held stocks
8. **Advanced Charting**
    - Candlestick charts for individual stocks
    - Technical indicators (RSI, MACD, Bollinger Bands)
9. **Compare Portfolios**
    - Compare performance across your portfolios
    - Public sharing links for portfolio comparison
10. **Mutual Fund Support**
    - NAV tracking for mutual funds
    - SIP tracking with folio numbers
11. **Crypto Portfolio**
    - Support cryptocurrency holdings
    - Integrate with CoinGecko/CoinMarketCap APIs
12. **Multi-Currency Support**
    - Track international investments
    - Automatic currency conversion

Long-term / Advanced Features

13. **Backtesting Engine**
    - Test investment strategies on historical data
    - "What if" scenarios
14. **Robo-Advisor Mode**
    - Fully automated portfolio management
    - Based on risk profile and goals

15. **Direct Broker Integration**
    - OAuth with Zerodha Kite, Groww, Upstox
    - Execute trades directly from app
16. **Real-time WebSocket Feed**
    - Live price updates without refresh
    - Streaming portfolio value
17. **Mobile Application**
    - React Native or Flutter
    - Push notifications for alerts
18. **Option Chain Analysis**
    - For F&O traders
    - Greeks calculation (Delta, Gamma, Theta, Vega)
19. **Tax Report Generator**
    - ITR-ready capital gains statement
    - Broker-wise P&L summary
20. **Family Portfolio View**
    - Aggregate multiple family members' portfolios
    - Combined household wealth tracking

📊 Feature Count Summary

Category	Implemented	Partial	Not Started
Authentication	9	0	3
Portfolio	7	1	0
Analytics	8	0	0
Rebalancing	5	0	0
Dividends	3	0	0
Tax	3	0	1
Export	3	0	0
AI	4	0	0
Notifications	5	2	1
UI/UX	7	1	0
Total	**54**	**4**	**11**

Completion Rate: ~82% of core features fully implemented

Technical Stack Summary

Backend

- **Framework**: FastAPI (Python)
- **Database**: PostgreSQL with SQLAlchemy ORM
- **Auth**: JWT with bcrypt password hashing
- **Email**: Brevo SMTP
- **Stock Data**: Alpha Vantage API

Frontend

- **Framework**: Next.js 14 (App Router)
- **Styling**: TailwindCSS
- **State**: Zustand with persist middleware
- **Charts**: Recharts
- **UI**: Custom components + SweetAlert2

> ***Note**: This audit represents a point-in-time snapshot. Feature status may change as development progresses.*